Predicting Chronic Kidney Disease



PG Group 13

Mohamed Elgazar

Vishwas More

Aisan Mojahedi

Gül Öner

Afzal Patel

Pratham Prashant Shah

github.com/elgazar24/f21dl-2026-27-PG-group-13

Table of Contents

1 Introduction **1**

2 Related Work **1**

3 Dataset Description and Analysis **1**
- 3.1 Dataset Overview . . . 1
- 3.2 Data Cleaning and Preprocessing . . . 2
- 3.3 Feature Engineering and Selection . . . 2
- 3.4 Clustering . . . 2

4 Experimental Setup **3**
- 4.1 Evaluation Protocol . . . 3
- 4.2 Classical Models . . . 3
- 4.3 Neural Networks . . . 4

5 Results **5**
- 5.1 Model Results . . . 5
- 5.2 Further Analyses . . . 5

6 Discussion **6**
- 6.1 Findings and Research Questions . . . 6
- 6.2 Relation to Prior Work . . . 6
- 6.3 Practical Implications . . . 6
- 6.4 Limitations and Future Work . . . 6

7 Conclusion **6**

References **iii**

Glossary **iii**

Member Contributions **iii**

1 Introduction

Chronic kidney disease (CKD) is common and growing. In 2023 it affected around **14% of adults** worldwide, about 788 million people, up from 378 million in 1990.[1] Its early stages usually have no symptoms, so many people do not know they have it until their kidneys are already damaged. This is one reason it was the **ninth leading cause of death** in 2023, with 1.5 million deaths.[1] *Machine learning* can help here. Standard blood and urine tests, like creatinine, haemoglobin and albumin, are already part of normal check-ups, and a model trained on these values could flag likely CKD cases cheaply. On the tabular data we compare three classical classifiers (kNN, Naive Bayes and Random Forest) with a multilayer perceptron (MLP); on the image data we train a convolutional neural network (CNN). The project asks **four questions**: which classical model predicts CKD best; which features matter most and whether the engineered ones help; whether unsupervised *clustering* groups patients in a way that matches the CKD diagnosis; and whether the MLP beats the best classical model, and how the CNN does on the images.

2 Related Work

Before starting, we reviewed earlier work on the tabular CKD dataset to decide where we could do something different. Polat et al.[2] used an SVM with feature selection. Qin et al.[3] compared six models and filled missing values with kNN imputation. Two things stood out. Both papers focus mainly on accuracy, which can hide how well the smaller class is predicted. It is also not clear whether the gaps were filled using all the data, including the test set, which can make results look better than they really are. We split the data before any preprocessing. We evaluate with macro F1. We also use clustering to see whether the data has any natural structure before relying on the labels.

Unlike these studies, we also have an image dataset. Here we looked at Islam et al.,[4] who compared CNNs (VGG16, ResNet, Inception) with vision transformers on about 12,400 kidney CT scans (normal, cyst, stone, tumour). In their study, CNNs worked well on kidney CT images, and the best transformer was only slightly better. Since CNNs are simpler to build and train, we use a CNN for our images.

3 Dataset Description and Analysis

3.1 Dataset Overview

TODO: DS1 and image data: source, license, features, statistics, peculiarities, why chosen.

3.2 Data Cleaning and Preprocessing

TODO: Steps and why, fit on train only.

3.3 Feature Engineering and Selection

TODO: New features, selection method, selected features.

3.4 Clustering

TODO: Methods, choice of k, results vs labels.

4 Experimental Setup

4.1 Evaluation Protocol

TODO: Split, cross-validation, metrics.

4.2 Classical Models

4.2.1 k-Nearest Neighbours

TODO: k, distance, why scaling matters.

4.2.2 Naive Bayes

TODO: Variant and its assumptions.

4.2.3 Random Forest

TODO: Trees, depth, feature importance.

4.3 Neural Networks

4.3.1 Multilayer Perceptron (MLP)

TODO: Layers, activation, optimiser, early stopping.

4.3.2 Convolutional Neural Network (CNN)

TODO: Architecture, augmentation, training.

5 Results

5.1 Model Results

TODO: Scores, confusion matrices, learning curves, all models compared, hypotheses supported or not.

5.2 Further Analyses

TODO: Extra experiments and why.

6 Discussion

6.1 Findings and Research Questions

TODO: Why the results happened.

6.2 Relation to Prior Work

TODO: Comparison with literature.

6.3 Practical Implications

TODO: Realism, reliability, risks.

6.4 Limitations and Future Work

TODO: Limitations and next steps.

7 Conclusion

TODO: Short summary, no numbers.

References

1. GBD 2023 CKD Collaborators. Global, regional, and national burden of chronic kidney disease in adults, 1990–2023. *Lancet*, 406, 2461–2482, 2025.
2. Polat, H. et al. Diagnosis of chronic kidney disease based on support vector machine by feature selection methods. *J. Med. Syst.*, 41, 2017.
3. Qin, J. et al. A machine learning methodology for diagnosing chronic kidney disease. *IEEE Access*, 8, 20991–21002, 2020.
4. Islam, M.N. et al. Vision transformer and explainable transfer learning models for auto detection of kidney cyst, stone and tumor from CT-radiography. *Sci. Rep.*, 12, 11440, 2022.

Glossary

Albumin	A protein healthy kidneys keep in the blood; in urine it is a sign of kidney damage.
CKD	A long-term condition in which the kidneys gradually lose their ability to filter the blood.
Creatinine	A waste product the kidneys filter out of the blood; high levels mean poor filtering.
Cyst	A fluid-filled sac that forms in the kidney and is usually harmless.
Feature selection	Keeping only the features that are most useful for prediction and dropping the rest.
Haemoglobin	The protein in red blood cells that carries oxygen; often low in people with CKD.
kNN imputation	Filling a missing value with the value taken from the most similar patients in the dataset.
Macro F1	The average of the two classes' F1 scores, so the smaller class counts equally.
Stone	A hard lump of minerals that builds up inside the kidney.
SVM	A model that finds the boundary that best separates the two classes.
Tumour	An abnormal growth of cells in the kidney, which may be harmless or cancerous.
Vision transformer	An image model that uses attention instead of convolutions to process the image.

Member Contributions

TODO: One line per member.